

DMDW Important Questions for MSE 2

1.What is classification and example. (Unit 4)

Classification (Definition)

Classification is the process of assigning data objects to one of the predefined classes or categories. It involves learning a function that maps input data (attributes) to a target class label.

In simple terms:

👉 It predicts **which category a new data belongs to** based on past data.

Example

From your notes:

- A **bank loan officer** classifies customers as **risky or safe** based on their data.
 - A **marketing manager** predicts whether a customer will **buy a computer or not**.
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One-line exam answer (if needed)

Classification is a data mining technique used to assign objects to predefined classes, such as classifying emails as spam or not spam.

2.What is support and confidence, why it is used and how to compute it? (Unit 3)

Support and Confidence (Association Analysis)

1. What is Support?

Support measures how frequently an itemset appears in the entire dataset. It tells **how common a rule is**.

$$\text{Support } s(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{N}$$

- $(\sigma(X \cup Y))$: number of transactions containing both X and Y
- (N) : total number of transactions

👉 Example: If {Milk, Bread} appears in 3 out of 10 transactions → support = $3/10 = 0.3$ (30%)

2. What is Confidence?

Confidence measures how often Y appears in transactions that contain X. It shows **the reliability of the rule**.

$$[\text{Confidence } c(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)}]$$

👉 Example: If 3 out of 5 transactions with Bread also contain Butter → confidence = $3/5 = 0.6$ (60%)

3. Why are Support and Confidence used?

- **Support**
 - Removes unimportant or rare rules
 - Ensures the rule is not due to chance
- **Confidence**
 - Measures strength/reliability of the rule
 - Higher confidence ⇒ stronger relationship

👉 Together, they help find **useful and meaningful association rules** from large datasets

4. How to Compute (Example from your file)

From dataset (5 transactions):

Rule: {Milk, Diaper} → {Beer}

- $(\sigma(\text{Milk, Diaper, Beer}) = 2)$
- Total transactions $(N = 5)$
- $(\sigma(\text{Milk, Diaper}) = 3)$

Support:

[
 $2/5 = 0.4 = 40\%$
]

Confidence:

[
 $2/3 = 0.67 = 67\%$
]

Exam-ready short answer

Support is the proportion of transactions containing both X and Y, while confidence is the probability that Y occurs given X. They are used to measure the strength and usefulness of association rules.

3. Hunts Algorithm (Unit 4)

Hunt's Algorithm (Decision Tree Induction)

Hunt's Algorithm is a **recursive method** used to construct a decision tree by splitting data into smaller and purer subsets.

Working of Hunt's Algorithm

Let D_t be the set of training records at node t :

1. **If all records in D_t belong to the same class**
→ Make node t a **leaf node** with that class label
 2. **If records belong to different classes**
→ Select an attribute test (best split)
→ Partition the data into smaller subsets
 3. **Repeat recursively**
→ Apply the same steps to each subset until all nodes become pure
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Key Idea

👉 “Divide the data again and again until each group contains only one class.”

Example (Simple Idea)

Suppose you are classifying students as **Pass/Fail**:

- If all students in a node are **Pass** → leaf = **Pass**
- If mixed → split using attributes like **Attendance, Marks**
- Continue splitting until classification is clear

Important Points

- It is **top-down and recursive**
- Produces a **decision tree**
- Stops when data becomes **pure (same class)**

Exam-ready Answer (Short)

Hunt's Algorithm is a recursive decision tree induction method that partitions training data into smaller subsets until each subset contains records of a single class, forming leaf nodes.

4.General Approach to solve classifical problem (Unit 4)

General Approach to Solve a Classification Problem

A classification task follows a structured process—from raw data to prediction. Based on your notes, here's the standard approach:

1. Data Collection

Gather the dataset with:

- Input attributes (features)
- Class labels (target/output)

👉 Example: Student data with attributes like attendance, marks, etc., and class label = Pass/Fail

2. Data Preprocessing

Prepare the data:

- Handle missing values
 - Remove noise
 - Normalize/transform data if needed
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3. Split the Dataset

Divide data into:

- **Training set** → used to build the model
 - **Test set** → used to evaluate performance
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4. Model Construction (Learning Phase)

Apply a classification algorithm such as:

- Decision Tree
- Rule-based classifier
- k-Nearest Neighbor
- Bayesian classifier

The model learns a function:

[
 $f(x) \rightarrow y$
]
(mapping attributes to class labels)

5. Model Evaluation

Use test data and evaluate using metrics like:

- Accuracy
 - Error rate
 - Confusion matrix
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6. Prediction (Deployment Phase)

Use the trained model to:

- Predict class labels for new/unseen data
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Key Idea

👉 Learn from known data → build model → test → predict unknown data

Exam-ready Answer (Short)

The general approach to classification involves data collection, preprocessing, splitting into training and test sets, building a model using classification algorithms, evaluating it using metrics like accuracy, and finally using it to predict class labels for new data.

5. Error rate, Accuracy rate and Confusion matrix (Unit 4)

Confusion Matrix, Accuracy, and Error Rate

1. Confusion Matrix

A **confusion matrix** is a table used to evaluate the performance of a classification model by comparing **actual vs predicted values**.

From your notes, the matrix is:

Actual \ Predicted	Pass	Fail
Pass	8	2
Fail	3	7

- 8 → correctly predicted Pass (True Positive)
- 2 → Pass predicted as Fail (False Negative)
- 3 → Fail predicted as Pass (False Positive)
- 7 → correctly predicted Fail (True Negative)

2. Accuracy Rate

Accuracy measures how many predictions are **correct** out of total predictions.

$$\begin{aligned}\text{Accuracy} &= \frac{TP + TN}{Total} \\ &= \frac{8 + 7}{8 + 2 + 3 + 7} = \frac{15}{20} = 75\%\end{aligned}$$

👉 Accuracy = 75%

3. Error Rate

Error rate measures how many predictions are **wrong**.

$$\begin{aligned}\text{Error Rate} &= \frac{FP + FN}{Total} \\ &= \frac{3 + 2}{20} = \frac{5}{20} = 25\%\end{aligned}$$

👉 Error Rate = 25%

Key Relationship

$$\text{Accuracy} + \text{Error Rate} = 1$$

Exam-ready Answer (Short)

A confusion matrix is a table showing actual vs predicted classifications. Accuracy is the ratio of correct predictions to total predictions, while error rate is the ratio of incorrect predictions to total predictions.